PHASE II AUTOMATION ROADMAP

ALLIES CONNECT - GROUP 4

Vision

The vision of the Phase II Automation Roadmap is to utilize three core components that encapsulate the sponsor's needs and key takeaways from the comparative analysis. All automation features will be implemented, but there will be an option for providers to opt out in the effect issues relating to that specific provider arise. Administrator and provider oversight will be incorporated for workflows that flag, ingest, or classify data to make sure that data is being validated. To ensure that these features are modular and at no cost, each feature will be independently implemented, developed, tested, integrated, and loosely coupled while using free-tier or Kennesaw State University-provided resources.

Recommended Approach

Volunteer Reminders and Scheduling

The automated volunteer reminders and scheduling feature should be implemented using Nodemailer and node-cron. With these two libraries, the system will be able to send emails to volunteers when they have registered for a shift, when they need to be reminded of an upcoming shift, and when they are asked for input based on their experience at the shift. Volunteers will be reminded forty-eight hours via email before their shift. Email templates will be stored in the existing database, and a new page will be created for shift feedback. The only new external dependency when implementing this feature is node-cron. This feature has already been partially implemented within Phase I. Phase II development will focus on the post-shift feedback email and workflow.

Resource Availability Verification

The automated resource availability verification feature should also be implemented using Nodemailer and node-cron. These two libraries allow the system to send scheduled emails to providers to verify the availability of resources that they supply. These emails will be scheduled to occur every thirty days. An additional workflow will need to be created to route the provider from their email to the resource update page if they need to edit the data. The only new external dependency when implementing this feature is node-cron.

Stale Data Detection

The automated stale data detection feature should also be implemented using Nodemailer and node-cron. These libraries allow the system to send scheduled emails to providers and administrators, if needed, to notify providers that their resource information has not been updated for ninety days. The provider will need to select whether the resource is active or not within the email. If they respond yes, the last updated element for the resource or event in their respective databases will be updated to the time at which they responded yes. If they select no, the provider will be brought to the resource or event page to edit or remove the data. If the provider does not respond within seven days, the event or resource will be flagged as stale and will not be visible to public users. This system requires a few additional workflows similar to the resource availability verification feature, but an additional workflow will be needed for flagging resources and events and deactivating them. The only new external dependency when implementing this feature is node-cron.

Calendar and Event Data Ingestion

The automated calendar and event data ingestion should be implemented using ical.js or ics.js within the Node.js backend of the application. This feature will require a new database to hold the calendar or data feed URLs. Node-cron will be used to schedule URL feed reading. Duplicate detection logic will be added to the backend to ensure data is not being added that already exists. The ingestion service will need to be designed so that any failures to pull data does not impact the website. If a failure is detected, an event needs to be triggered to notify administrators of the issue so they can further investigate it. Additionally, a workflow will need to be created to notify providers when events are created via the URL feed. The only new external dependency when implementing this feature is node-cron for scheduling the feed pulling.

Potential Pilot Partners

Pilot testing with a restricted group of nonprofit organizations needs to be performed before all of these features are fully implemented into the system. It is important that this is done so that missed edge cases and unexpected failures are caught and do not cause a full shutdown of the live web service. Pilot partners need to be carefully selected based on a set of criteria. First, the organization needs to have an existing relationship with Allies Way, Inc. Second, they are already registered on the Allies Connect platform. Third, they are capable and willing to provide accurate and honest feedback for the benefit of the Phase II development. Finally, the organization utilizes Google Calendar or some other ICS-based scheduling platform data ingestion testing.

Proposed Implementation Timeline

Stage (Timeline)	Priority Tier	Features	Technical Approach	Rationale
Stage 1 (Months 1-2)	Must Have	Volunteer Reminders and Scheduling (partially implemented) Resource Availability Verification Stale Data Detection	Nodemailer, node-cron	Low cost, low risk, low effort, high benefit
Stage 2 (Months 3-4)	Should Have	Calendar and Event Data Ingestion	ical.js, ics.js, node-cron, Nodemailer	Low cost, mid risk, mid effort, high benefit
Future Phase	Nice to Have	Data Classification	Free-tier LLM API resources, key-word directory	Mid cost, mid risk, mid effort, high benefit
N/A	Not Recommended	Resource and Event Updates	Paid LLM API and scraping resources	Mid cost, high risk, high effort, high benefit